The Exact Noncompactness Norm of the Calkin Representation

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Status

This packet records a candidate partial result, likely valid. It sharpens the previous lane-5 packet on Remark 1 after Theorem 2.1 of Boedihardjo–Johnson, arXiv:1301.0054. The source question asks whether their Calkin representation theorem remains true without assuming that X has the bounded compact approximation property (BCAP).

Source Context

Boedihardjo and Johnson define, for each infinite-dimensional Banach space Y, the ultrapower space

$$\widehat{Y} = \{(y_\alpha^*)_{\alpha,\mathcal{U}} \in (Y^*)^{\mathcal{U}} : w^*\text{-}\lim_{\alpha,\mathcal{U}} y_\alpha^* = 0\},$$

where α ranges over finite-dimensional subspaces of a fixed infinite-dimensional Banach space X. If $T \in B(X,Y)$, they define $\widehat{T} \in B(\widehat{Y}, \widehat{X})$ by

$$(y_\alpha^*)_{\alpha,\mathcal{U}} \mapsto (T^* y_\alpha^*)_{\alpha,\mathcal{U}}.$$

They observe that $\widehat{K} = 0$ for compact K. Their Theorem 2.1 states that, if X has the λ-BCAP, then

$$f : B(X)/K(X) \to B(\widehat{X}), \qquad \dot{T} \mapsto \widehat{T},$$

is a norm-one $(\lambda + 1)$-embedding.

ultrapower (see e.g., [2, Chapter 8]) of Y^* with respect to U. (The ultrafilter U and the directed set Λ_0 do not depend on Y.) If $(y^*_\alpha)_{\alpha\in\Lambda_0}$ is a bounded net in Y^*, then its image in $(Y^*)^U$ is denoted by $(y^*_\alpha)_{\alpha,U}$. Consider the (complemented) subspace

$$\widehat{Y} := \left\{ (y^*_\alpha)_{\alpha,U} \in (Y^*)^U : w^*\text{-}\lim_{\alpha,U} y^*_\alpha = 0 \right\}$$

of $(Y^*)^U$. Here $w^*\text{-}\lim\limits_{\alpha,U} y^*_\alpha$ is the w^*-limit of $(y^*_\alpha)_{\alpha\in\Lambda_0}$ through U, which exists by the Banach-Alaoglu Theorem.

Whenever $T \in B(X,Y)$, we can define an operator $\widehat{T} \in B(\widehat{Y}, \widehat{X})$ by sending $(y^*_\alpha)_{\alpha,U}$ to $(T^*y^*_\alpha)_{\alpha,U}$. Note that if $K \in K(X,Y)$ then $\widehat{K} = 0$, where $K(X,Y)$ denotes the space of all compact operators in $B(X,Y)$.

Theorem 2.1. *Suppose that X has the λ-BCAP. Then the operator $f : B(X)/K(X) \to B(\widehat{X})$, $\dot{T} \mapsto \widehat{T}$, is a norm one $(\lambda+1)$-embedding into $B(\widehat{X})$ satisfying*

$$f(\dot{I}) = I \text{ and } f(\dot{T}_1\dot{T}_2) = f(\dot{T}_2)f(\dot{T}_1), \qquad T_1, T_2 \in B(X).$$

Proof. It is easy to verify that f is a linear map, $f(\dot{I}) = I$, and $f(\dot{T}_1\dot{T}_2) = f(\dot{T}_2)f(\dot{T}_1)$ for $T_1, T_2 \in B(X)$. If $T \in B(X)$, then clearly $\|f(\dot{T})\| \le \|T\|$, and thus we also have $\|f(\dot{T})\| \le \|T\|_e$. Hence $\|f\| \le 1$. It remains to show that f is a $(\lambda+1)$-embedding (i.e., $\inf\limits_{\|T\|_e>1} \|f(\dot{T})\| \ge (\lambda+1)^{-1}$).

To do this, let $T \in B(X)$ satisfy $\|T\|_e > 1$. Since X has the λ-BCAP, we can find a net of operators $(S_\alpha)_{\alpha\in\Lambda_0} \subset K(X)$ converging strongly to I such that $\sup\limits_{\alpha\in\Lambda_0} \|S_\alpha\| \le \lambda$. Then $\|T^*(I-S_\alpha)^*\| = \|(I-S_\alpha)T\| \ge \|T\|_e > 1$, $\alpha \in \Lambda_0$. Thus, there exists $(x^*_\alpha)_{\alpha\in\Lambda_0} \subset X^*$ such that $\|x^*_\alpha\| = 1$ and $\|T^*(I-S_\alpha)^*x^*_\alpha\| > 1$ for $\alpha \in \Lambda_0$.

Note that for every $x \in X$,

$$\limsup_{\alpha\in\Lambda_0} |\langle (I-S_\alpha)^*x^*_\alpha, x\rangle| = \limsup_{\alpha\in\Lambda_0} |\langle x^*_\alpha, (I-S_\alpha)x\rangle| \le \limsup_{\alpha\in\Lambda_0} \|(I-S_\alpha)x\| = 0,$$

and so the net $((I-S_\alpha)^*x^*_\alpha)_{\alpha\in\Lambda_0}$ converges in the w^*-topology to 0. By the construction of U, this implies that

Immediately after the proof, the authors ask whether Theorem 2.1 remains true without the BCAP hypothesis.

It follows that $\|f(\dot{T})\| \ge (1+\lambda)^{-1}$ whenever $\|T\|_e > 1$. $\square$

Remark 1. We do not know whether Theorem 2.1 is true without the hypothesis that X has the BCAP.

Remark 2. The embedding in Theorem 2.1 is an isometry if the approximating net can be chosen so that $\|I - S_\alpha\| = 1$ for every α. This is the case if, for example, the space X has a 1-unconditional basis. However, we do not know whether the embedding is an isometry if $X = L_p(0,1)$ with $p \ne 2$.

If N is a subset of Y^*, then we can define a subset N' of $\widehat{Y}$ by

Result

For $T \in B(X)$ and a finite-dimensional subspace $E \subset X$, write $Q_E : X \to X/E$ for the quotient map and set

$$\kappa_X(T) := \inf_{E\subset X,\ \dim E<\infty} \|Q_E T\|.$$

This is the limiting Kolmogorov noncompactness quantity of the set $T(B_X)$.

Theorem 1. *Let X be an infinite-dimensional Banach space and use the Boedihardjo–Johnson ultrafilter on the directed set of finite-dimensional subspaces of X. Then, for every $T \in B(X)$,*

$$\|\widehat{T}\| = \kappa_X(T) = \inf_{E\subset X,\ \dim E<\infty} \|Q_E T\|.$$

Proof Intuition

The lower bound is the construction from the previous packet: for each finite-dimensional E, take a norm-one functional in $E^\perp$ on which T^* almost attains $\|T^*|_{E^\perp}\| = \|Q_E T\|$. These functionals are weak-star-null along the tail ultrafilter, so they give a test vector in $\widehat{X}$.

The new point is the reverse inequality. Any vector of $\widehat{X}$ is represented by a bounded weak-star-null family (x^*_α). Fix a finite-dimensional $F \subset X$. Since weak-star convergence on F is norm convergence in F^*, the restrictions $x^*_\alpha|_F$ are small on an ultrafilter-large set. Equivalently, x^*_α is close to $F^\perp$. Replacing x^*_α by a nearby element of $F^\perp$ forces $T^* x^*_\alpha$ to be bounded by $\|T^*|_{F^\perp}\| = \|Q_F T\|$. Taking the infimum over F gives the upper bound.

Proof. Let Λ_0 be the directed set of finite-dimensional subspaces of X, ordered by inclusion, and let $\mathcal{U}$ be an ultrafilter containing every tail

$$\{\alpha \in \Lambda_0 : \alpha \supset \alpha_0\}.$$

For $E \in \Lambda_0$, duality gives

$$\|Q_E T\| = \|(Q_E T)^*\| = \|T^*|_{E^\perp}\|.$$

The function $E \mapsto \|Q_E T\|$ is decreasing on Λ_0, so its $\mathcal{U}$-limit is its infimum, namely $\kappa_X(T)$.

We first prove $\|\widehat{T}\| \geq \kappa_X(T)$. Fix $\varepsilon > 0$. For every $E \in \Lambda_0$, choose $x^*_E \in E^\perp$ with $\|x^*_E\| = 1$ and

$$\|T^* x^*_E\| \geq \|T^*|_{E^\perp}\| - \varepsilon = \|Q_E T\| - \varepsilon.$$

The family $(x^*_E)_{E,\mathcal{U}}$ belongs to $\widehat{X}$, because for each $x \in X$, the tail $E \supset \operatorname{span}\{x\}$ lies in $\mathcal{U}$, and on that tail $x^*_E(x) = 0$. Therefore

$$\|\widehat{T}\| \geq \lim_{E,\mathcal{U}} \|T^* x^*_E\| \geq \lim_{E,\mathcal{U}} \|Q_E T\| - \varepsilon = \kappa_X(T) - \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives the lower bound.

For the upper bound, let $\xi = (x^*_E)_{E,\mathcal{U}} \in \widehat{X}$ with $\|\xi\| \leq 1$. Fix a finite-dimensional subspace $F \subset X$ and $\varepsilon > 0$. Since $w^*\text{-}\lim_{E,\mathcal{U}} x^*_E = 0$, and since F is finite dimensional,

$$\lim_{E,\mathcal{U}} \|x^*_E|_F\|_{F^*} = 0.$$

For E in an ultrafilter-large set, choose $z^*_E \in F^\perp$ with

$$\|x^*_E - z^*_E\| \leq \|x^*_E|_F\| + \varepsilon.$$

This is possible because the quotient $X^*/F^\perp$ is isometric to F^* by restriction. Hence, on an ultrafilter-large set, $\|x_E^* - z_E^*\| \le 2\varepsilon$. It follows that

$$\|T^* x_E^*\| \le \|T^* z_E^*\| + 2\varepsilon\|T\| \le \|T^*|_{F^\perp}\| \|z_E^*\| + 2\varepsilon\|T\|.$$

Also $\|z_E^*\| \le \|x_E^*\| + 2\varepsilon$. Taking $\mathcal{U}$-limits and using $\lim_{E,\mathcal{U}} \|x_E^*\| = \|\xi\| \le 1$, we obtain

$$\|\widehat{T}\xi\| \le \|T^*|_{F^\perp}\|(1+2\varepsilon) + 2\varepsilon\|T\| = \|Q_F T\|(1+2\varepsilon) + 2\varepsilon\|T\|.$$

Letting $\varepsilon \downarrow 0$, and then taking the infimum over finite-dimensional $F \subset X$, gives

$$\|\widehat{T}\xi\| \le \kappa_X(T).$$

Since this holds for every $\xi \in \widehat{X}$ with $\|\xi\| \le 1$, we have $\|\widehat{T}\| \le \kappa_X(T)$. This completes the proof. $\square$

Corollary 1. *The Boedihardjo–Johnson representation is injective modulo compact operators for every infinite-dimensional Banach space X.*

Proof. The quantity $\kappa_X(T)$ is zero if and only if T is compact. Indeed, if $\kappa_X(T) = 0$, then for every $\varepsilon > 0$, the set $T(B_X)$ lies within ε of a finite-dimensional subspace of X. Since $T(B_X)$ is bounded, this makes $T(B_X)$ totally bounded. Conversely compact sets are approximated by finite-dimensional spans of finite nets. The theorem now gives $\widehat{T} = 0$ if and only if T is compact. $\square$

Reduction of the Remaining Open Problem

The theorem identifies exactly what the ultrapower representation measures. Thus Theorem 2.1 without BCAP is equivalent to asking whether, for the relevant Banach space X, there is a constant $C_X < \infty$ such that

$$\|T\|_e \le C_X\, \kappa_X(T) \qquad (T \in B(X)).$$

The BCAP proof in Boedihardjo–Johnson proves precisely this estimate with $C_X = \lambda + 1$ when X has the λ-BCAP. Without BCAP, the representation remains faithful, but the unresolved issue is whether the essential norm and the Kolmogorov noncompactness norm are uniformly comparable on $B(X)/K(X)$.

Remark 2 is similarly reframed: for $X = L_p(0,1)$, $p \ne 2$, the representation is isometric exactly when

$$\|T\|_e = \kappa_X(T) \qquad (T \in B(X)).$$

Novelty and Literature Check

The run indexes were searched for arXiv:1301.0054, Boedihardjo–Johnson, Calkin, BCAP, essential norm, and measure/Kolmogorov noncompactness phrases. The only direct run hit was the prior lane-5 injectivity packet. Bounded web searches for exact phrases around Theorem 2.1, the BCAP-free Calkin representation, and essential norm versus measure of noncompactness did not surface a later paper resolving Remark 1. Novelty confidence is moderate: the norm formula is natural once one looks at the quotient norms, but it is not recorded in the run memory and was not visible in the bounded search.

Limitations

This packet does not settle whether the estimate $\|T\|_e \leq C_X \kappa_X(T)$ holds for every Banach space, nor does it produce a space and operators for which the ratio $\|T\|_e/\kappa_X(T)$ is unbounded. It should therefore be treated as a sharp reduction and a genuine strengthening of the injectivity result, not as a full solution of Remark 1.

Human Review Recommendation

Reviewers should check the upper-bound argument in the theorem, especially the use of finite-dimensionality of F to turn weak-star convergence into norm convergence of restrictions to F, and the isometric identification $X^*/F^\perp \simeq F^*$.

References

[1] M. T. Boedihardjo and W. B. Johnson, *On Mean Ergodic Convergence in the Calkin Algebras*, arXiv:1301.0054.